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ORIGINAL ARTICLE

Temporal and spatial effects of environmental variables on crustacean communities in Çardak Lagoon (Turkish Straits System) under the influence of domestic pollution

Abdullah Suat Ateş[1] · Ayşegül Mülayim[2] · Ahmet Kerem Bakir[3] · Seçil Acar[1] · Yeşim Büyükateş[1] · Ertan Dağlı[3] · Alper Doğan[3] · Sinan Uzundumlu[4] · Çetin Kedioğlu[5]

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Abstract

This study presents the temporal and spatial effects of domestic pollution on crustacean assemblages in a protected area, Çardak Lagoon (Turkish Straits System). The samplings were performed from sandy and mud (silt + clay) bottoms between 1 and 1.8 m depths in October 2018, February, April, and June 2019. A total of 6920 specimens and 26 species (1 Isopoda, 1 Tanaidacea, 2 Mysidacea, 3 Cumacea, 5 Decapoda, 14 Amphipoda) belonging to crustaceans was found. Highest crustacea abundances were in autumn and winter. Considering the environmental variables, $NO_2 + NO_3$ (mg L^{-1}) and COD (mg L^{-1}) levels in the water had the highest correlations with abundance for the sampling periods. According to sampling points, the highest correlation value was found between anionic detergent content in the water and the abundance. Crustacean community structure was modified by environmental variables of anthropogenic origin at a higher level. Oxygen, oxygen reduction potential, $NO_2 + NO_3$, total nitrogen, chemical oxygen demand and anionic surfactant levels in water and sediment sand content % were major environmental variables affecting the crustacean abundance in the study area temporally.

Keywords Crustacea community · Environmental variables · Domestic pollution · Çardak lagoon · Turkish Straits System

Introduction

Domestic pollution exhibits the most common anthropogenic and negative effects on marine benthic communities (Wear and Tanner 2007; Acquavita et al. 2015). Eutrophication observed in nutrient-rich environments is a major problem in semi-closed lagoons. Domestic discharges cause excessive nutrient and organic matter accumulation in coastal lagoons (Magni et al. 2005). In ecotones such as lagoons, oxygen level in water decreases considerably and hydrogen sulfide release increases by excessive eutrophication. Increasing organic matter amount due to discharges causes sharp daily and seasonal changes in water quality variables (Lardicci et al. 2001). Therefore, a mass death of macrozoobenthos can be observed (Bachelet et al. 2000). Macrozoobenthic communities react relatively quickly to anthropogenic and natural stresses (Pearson and Rosenberg 1978). Depending on the dystrophic conditions in the discharge areas where domestic waste reaches the coasts, benthic species initially increase quantitatively and qualitatively, while sensitive species disappear (Bat et al. 2001). As known, peracarid crustaceans are major components of soft sediment fauna and known to be good indicators of water and sediment quality. Similarly, detritivore peracarids are

✉ Abdullah Suat Ateş
asuatates@yahoo.com

1 Faculty of Marine Sciences and Technology Çanakkale, Onsekiz Mart University, Çanakkale 17100, Turkey

2 Faculty of Science Department of Biology, İstanbul University, İstanbul, Vezneciler 34134, Turkey

3 Faculty of Fisheries, Ege University, İzmir, Bornova 35100, Turkey

4 Hınıs Vocational School Atatürk University, Erzurum 25240, Turkey

5 Vocational School of Marine Technologies Çanakkale Onsekiz Mart University, Çanakkale 17100, Turkey

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important in the decomposition of organic matter in aquatic sediments (Jelassi et al. 2015). Changes in benthic community structure are defined depending on the degree of anthropogenic pollution (Arasaki et al. 2004). Anthropogenic pollution has negative effect on crustacean diversity, especially on amphipod species (Sánchez-Moyano et al. 2006).

In recent 30 years, descriptive studies on macrozoobenthic communities, including crustaceans in Mediterranean Lagoons, (Nicolaidou et al. 1988; Lardicci et al. 1997; Basset et al. 2006; Nicolaidou et al. 2006; Afli et al. 2008; Paraskevopoulou et al. 2015) and levels of disturbance due to environmental variables (Reizopoulou et al. 1996; Lardicci and Rossi 1998; Bachelet et al. 2000; Koutsoubas et al. 2000; Lardicci et al. 2001; Arvanitidis et al. 2005; Afli et al. 2009a, b; Foti et al. 2014; Ounifi-Ben Amor et al. 2017) were performed.

Considering the relevant literature, the relationships between crustacean assemblages and domestic environmental variables in Turkish lagoons were not comprehensively studied. Zoogeographic distributions of some crustacean species observed in six Turkish coastal lagoons were previously reported by Kocatas and Katağan (1983). Akbulut et al. (2002) conducted a study on crustacean assemblages and their ecology in Sarıkum Lagoon (Sinop coast, southern Black Sea). Recently, Özbek et al. (2016) performed a study concerning malacostracans of Karina Lagoon in the eastern Aegean Sea. Thus, this study was aimed to reveal both temporal and spatial effects of the domestic pollution and environmental conditions on the crustacean assemblages in Çardak Lagoon.

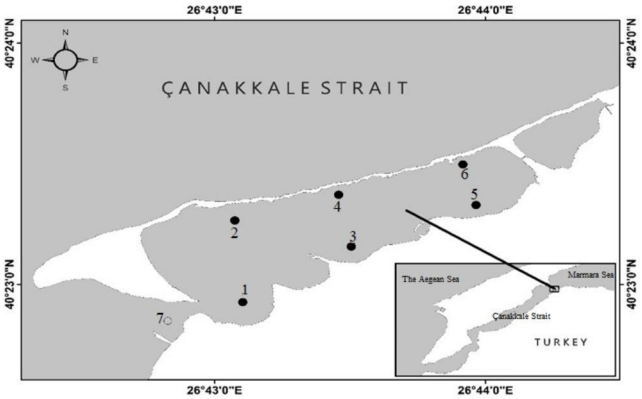


Fig. 1 Map of the study area. 1–7 sampling points in the Çardak Lagoon



Fig. 2 Views of the study area (Photograhed by A.S. Ateş)

Materials and methods

Water quality analysis

Water quality variables such as temperature, salinity, pH and dissolved oxygen solubility were measured in *situ* by means of a YSI 600 QS Multiprobe System (Yellow Springs Instruments) in sampling sites throughout the sampling periods. To determine the nutrient concentrations water samples were collected using a Nansen sampler of 5 L water. Samples for nutrients such as NO^{-}_{2}, NO^{-}_{3}, NH^{+}_{4}, PO^{-}_{4}-P and SiO^{-}_{2} were measured using a Jasco Brand UV Spectrophotometer based on chemical and biological analysis techniques of Strickland and Parsons (1972). Measurements were performed in various wavelengths in such 543 nm, 410 nm, 640 nm, 850 nm and 810 nm for NO^{-}_{2}, NO^{-}_{3}, NH^{+}_{4}, PO^{-}_{4}-P, SiO^{-}_{2}, respectively. For the determination of the chlorophyll-*a* concentrations, water samples were taken from each site via 5 L water sampler. Each sample was vacuum filtered in situ using a 47 mm GF/F filter, then wrapped by aluminum foil and kept frozen until analysis. The spectrophotometric analysis after 90% acetone extraction was performed according to APHA (1995). Total suspended solids (TSS) determination was performed gravimetrically after the water samples were filtered through GF/C filters (Clesceri et al. 1998). Chemical oxygen demand (COD) analysis was conducted using the open reflux method and standard methods for the examination of water and wastewater according to Eaton and Franson (2005). For the spectrophotometric determination of anionic surfactants, a standard procedure was conducted using methylene blue (APHA 1992).

Sediment sampling

Sediment material was sampled from 1 to 1.8 m depths of seven sampling points (GPS Coordinates: 40°23’14“ N, 26°43’30” E) (Figs. 1, 2).

The benthos samples were collected in three replicates in October 2018, February, April, and June 2019, using a metal

framed quadrat system of 100 × 100 cm by a SCUBA diver (Fig. 3). The material was fixed in neutralized 4% formaldehyde in 1 and 5 L plastic bottles.

Faunistic analysis

Benthos materials were washed and passed through a triple sieve system with mesh sizes of 0.5, 1, and 2 mm in the laboratory with pressurized water. Subsequently, crustaceans on the sieves were extracted at macro and micro level. Crustacean specimens were diagnosed in trinocular stereomicroscope based on taxonomic descriptions of Tattersall and Tattersall (1951), Ledoyer (1965), Riggio (1996), Dojiri and Sieg (1997), Bellan-Santini et al. (1982, 1989, 1993), and Falciai and Minervini (1996).

Bellan-Santini (1969) dominance index (di%) was used to calculate the crustacean dominance. The correlations between environmental variables and crustacean abundance were calculated according to Pearson coefficient correlation (r) in PAST 4.02 program. H′ (Shannon and Wiever), BENTIX (Simboura and Zenetos 2002), and TUBI (Çınar et al. 2015) indices were used to determine the ecological quality of the study area.

TUBI Index = Metric 1 + (5- Metric 2) / 2, TUBI = H′† + [0 G1%+3 × G2%+5 × G3% / 100] / 2.

In the formula, GI = the group of species sensitive to pollution, GII = the group of species that respond to stress and pollution, GIII = the group of hypoxia-tolerant species, GIV = the group of second-degree opportunistic species, GV = the group of opportunistic species of the first degree. In the calculation of metric 2, 3 ecological groups are considered. Group 1 (GI and GII), Group 2 (GIII), Group 3 (GIV and GV). Correlations between dominant species and environmental variables were found by Principal correspondence analysis (Jolliffe 2002).



Fig. 3 Samplings underwater (Photographed by Ç. Kedioğlu)

Analyses of sediment characteristics

A sediment core made of 393 cm^3 acrylic material was used for the organic matter and the particle size analyses in soft bottoms of the lagoon. A total of 28 core samples for both analyses were obtained from soft sediments at each sampling point in each sampling period. Particle size analyses of the sediment was performed according to Allen (1997). Organic matter (OM%) content was determined as the difference between the dry weight (80 °C, 24 h) of the sediment and the residue left after combustion at 450 °C for 2 h (Parker 1983).

Results

Water quality, nutrients, chl-*a*, TSS, COD, and anionic surfactants

Water quality variables followed the seasonal fluctuations. Temperature ranged between 7.57 and 27.34 ºC having the highest value in Summer 2019. Salinity changed between 20.17‰ and 24.40‰. pH followed typical sea water values ranging between 7.78 and 8.56. DO changed between 5.90 and 9.51 mg L^{-1}. NH_4 values were at very low levels (<0.01 mg L^{-1}) at all stations throughout the sampling period. PO_4-P values were in the range of 0.01–0.03 mg L^{-1}. While total phosphate values varied between 0.02 and 0.17 mg L^{-1}, the highest total nitrogen value was (0.99 mg L^{-1}). In terms of $NO_2 + NO_3$ values, the highest value (0.195 mg L^{-1}) was in autumn 2018. The highest chl-*a* value (12.85 µg L^{-1}) was found in summer 2019 and the highest TSS value (71.33 mg L^{-1}) was recorded in winter 2019. As a result of the correlation analysis, a significant negative linear relationship ($p < 0.001$) was recorded between chlorophyll-*a* and TSS values indicating that the load in the system was terrestrial rather than phytoplanktonic. COD values (mg L^{-1}) were higher in autumn and winter compared to other seasons. On the contrary, anionic surfactant values (mg L^{-1}) showed lower values in autumn and winter compared to other seasons. Mean values of all environmental variables throughout the year during the sampling period were presented in Table 1.

Granulometry

Sediment in the sampling area was composed of 71.59% sand, 20.16% gravel + shell, and 8.19% mud (clay + silt). While the highest sand ratio was at station 6 with a value of 92%, the highest mud (clay + silt) content (16.71%) and the highest gravel + shell ratio (25.64%) were observed at station 5 (Table 2).

Table 1 Water quality, nutrient, chl-*a*, suspended solids (TSS), chemical oxygen demand (COD), and anionic surfactant (AS) values measured in the Çardak Lagoon

Variable	Station	Mean ± SD	Min.	Max.	Variable	Station	Mean ± SD	Min.	Max.
T (°C)	St. 1	15.59 ± 6.77	8.46	24.75	pH	St. 1	7.97 ± 0.19	7.78	8.20
	St. 2	16.74 ± 8.07	7.86	27.34		St. 2	8.29 ± 0.08	8.18	8.37
	St. 3	15.61 ± 7.21	7.69	25.20		St. 3	8.18 ± 0.09	8.05	8.26
	St. 4	15.53 ± 6.61	7.96	24.07		St. 4	8.32 ± 0.09	8.23	8.43
	St. 5	15.89 ± 7.85	7.57	26.31		St. 5	8.25 ± 0.15	8.07	8.40
	St. 6	16.33 ± 7.88	7.68	26.54		St. 6	8.34 ± 0.20	8.14	8.56
	St. 7, ref.	16.19 ± 7.03	8.72	25.46		St. 7, ref.	8.25 ± 0.09	8.12	8.35
S (‰)	St. 1	21.91 ± 1.69	20.29	23.92	O_2 (mg L^{-1})	St. 1	8.18 ± 1.49	6.11	9.51
	St. 2	22.01 ± 1.37	20.78	23.92		St. 2	7.76 ± 1.19	6.85	9.50
	St. 3	21.79 ± 1.20	20.76	23.00		St. 3	7.21 ± 1.08	6.15	8.17
	St. 4	22.04 ± 1.37	20.77	23.69		St. 4	7.07 ± 1.01	5.99	8.27
	St. 5	22.04 ± 0.88	20.80	22.85		St. 5	7.18 ± 1.12	6.16	8.47
	St. 6	21.87 ± 0.84	20.74	22.74		St. 6	7.08 ± 0.81	5.90	7.69
	St. 7, ref.	22.04 ± 1.93	20.17	24.40		St. 7, ref.	7.76 ± 0.37	7.41	8.20
PO_4 (mg L^{-1})	St. 1	0.02 ± 0.012	0.01	0.03	NH_4 (mg L^{-1})	St. 1	0.01 ± 0.00	0.01	0.01
	St. 2	0.015 ± 0.006	0.01	0.02		St. 2	0.01 ± 0.00	0.01	0.01
	St. 3	0.013 ± 0.005	0.01	0.02		St. 3	0.01 ± 0.00	0.01	0.01
	St. 4	0.015 ± 0.006	0.01	0.02		St. 4	0.01 ± 0.00	0.01	0.01
	St. 5	0.01 ± 0.00	0.01	0.01		St. 5	0.01 ± 0.00	0.01	0.01
	St. 6	0.015 ± 0.01	0.01	0.03		St. 6	0.01 ± 0.00	0.01	0.01
	St. 7, ref.	0.01 ± 0.00	0.01	0.01		St. 7, ref.	0.01 ± 0.00	0.01	0.01
TP (mg L^{-1})	St. 1	0.048 ± 0.026	0.02	0.07	TN (mg L^{-1})	St. 1	0.480 ± 0.206	0.30	0.74
	St. 2	0.029 ± 0.013	0.02	0.04		St. 2	0.230 ± 0.126	0.10	0.37
	St. 3	0.026 ± 0.009	0.02	0.04		St. 3	0.427 ± 0.170	0.25	0.66
	St. 4	0.035 ± 0.018	0.02	0.05		St. 4	0.162 ± 0.047	0.10	0.20
	St. 5	0.051 ± 0.066	0.02	0.15		St. 5	0.498 ± 0.379	0.20	0.99
	St. 6	0.065 ± 0.073	0.02	0.17		St. 6	0.498 ± 0.379	0.20	0.99
	St. 7, ref.	0.052 ± 0.072	0.02	0.16		St. 7, ref.	0.262 ± 0.075	0.20	0.35
$NO_2 + NO_3$ (mg L^{-1})	St. 1	0.083 ± 0.04	0.04	0.13	SiO_2 (mg L^{-1})	St. 1	0.367 ± 0.241	0.15	0.60
	St. 2	0.071 ± 0.059	0.04	0.13		St. 2	0.563 ± 0.442	0.20	1.20
	St. 3	0.071 ± 0.059	0.02	0.13		St. 3	0.292 ± 0.225	0.05	0.55
	St. 4	0.036 ± 0.02	0.01	0.06		St. 4	0.575 ± 0.561	0.20	1.40
	St. 5	0.088 ± 0.074	0.03	0.20		St. 5	0.555 ± 0.632	0.17	1.50
	St. 6	0.088 ± 0.063	0.03	0.18		St. 6	0.186 ± 0.156	0.02	0.40
	St. 7, ref.	0.094 ± 0.041	0.05	0.14		St. 7, ref.	0.487 ± 0.477	0.20	1.20
Chl-*a* (μg L^{-1})	St. 1	3.64 ± 3.84	1.61	9.39	TSS (mg L^{-1})	St. 1	11.00 ± 4.62	6.80	17.60
	St. 2	2.64 ± 1.52	1.06	3.96		St. 2	8.07 ± 2.74	4.30	10.20
	St. 3	3.39 ± 2.27	1.09	5.75		St. 3	22.1 ± 22.5	3.80	54.40
	St. 4	2.96 ± 2.26	0.97	6.19		St. 4	9.35 ± 4.85	4.00	15.60
	St. 5	2.59 ± 2.13	1.31	5.76		St. 5	19.1 ± 24.8	3.20	56.00
	St. 6	4.47 ± 5.61	1.07	12.85		St. 6	28.4 ± 29.7	7.20	71.30
	St. 7, ref.	1.49 ± 0.44	1.05	1.98		St. 7, ref.	15.98 ± 8.36	7.70	24.40
COD (mg L^{-1})	St. 1	127.9 ± 52	76.00	198	AS (mg L^{-1})	St. 1	0.045 ± 0.022	0.02	0.07
	St. 2	152.1 ± 102.8	76.00	295		St. 2	0.027 ± 0.016	0.02	0.05
	St. 3	105.0 ± 55.2	40.00	158		St. 3	0.032 ± 0.014	0.02	0.05
	St. 4	87.5 ± 64.1	40.00	181		St. 4	0.040 ± 0.014	0.02	0.05
	St. 5	120.5 ± 111.8	40.00	277		St. 5	0.034 ± 0.011	0.02	0.05
	St. 6	148.3 ± 98.1	53.00	277		St. 6	0.035 ± 0.012	0.02	0.05
	St. 7, ref.	101.5 ± 79.6	40.00	207		St. 7, ref.	0.032 ± 0.016	0.02	0.05

Table 2 Precent granulometry ratios recorded at stations in the Çardak Lagoon

Particle type %								
Factor	Clay + silt	Very fine sand	Fine sand	Medium sand	Coarse sand	Very coarse sand	Gravel and shell	Coarse gravel and shell
St. 1	14.16	13.89	18.04	9.43	8.91	11.46	14.74	9.3
St. 2	3.17	10.96	15.93	19.57	16.49	15.72	10.47	7.65
St. 3	13.24	7.25	7.92	9.85	12.82	17.3	19.72	11.83
St. 4	5.27	20.08	43.68	7.1	8.91	6.42	4.12	4.36
St. 5	16.71	23.66	9.27	7.17	7.75	9.76	11.85	13.79
St. 6	3.29	13.67	43.94	25.06	6.16	3.91	2.59	1.38
St. 7, ref.	1.49	3.1	15.26	17.55	15.88	17.3	16.11	13.25
Mean	8.19 ± 6.27	13.23 ± 7.06	22.00 ± 15.32	13.67 ± 4.04	10.98 ± 4.08	11.69 ± 5.34	11.37 ± 6.24	8.79 ± 4.65

Table 3 The amount of organic matter in the water and sediment measured in the Çardak Lagoon sampling points

Period	Station	OMW (mg L^{-1})	OM %	Period	Station	OMW (mg L^{-1})	OM %
	St. 1	11	11.43		St. 1	11.6	8.9
	St. 2	10.8	2.73		St. 2	14.6	3.99
	St. 3	11.1	14.21		St. 3	12.8	14.13
Autumn	St. 4	11.2	4.38	Winter	St. 4	10.6	2.97
	St. 5	11	7.76		St. 5	12.2	7.53
	St. 6	10.9	2.69		St. 6	12.4	2.94
	St. 7, ref.	11.2	1.76		St. 7, ref.	11.2	1.4
	St. 1	10.6	10.14		St. 1	11.2	12.14
	St. 2	9.6	3.6		St. 2	11.6	1.82
	St. 3	11.4	16.88		St. 3	11.2	16.85
Spring	St. 4	10.0	4.08	Summer	St. 4	11.7	2.56
	St. 5	9.6	6.76		St. 5	12.0	9.6
	St. 6	10.0	2.73		St. 6	11.4	2.64
	St. 7, ref.	9.6	2.1		St. 7, ref.	11.3	1.69

OMW: organic matter in water, OM: organic matter in sediment

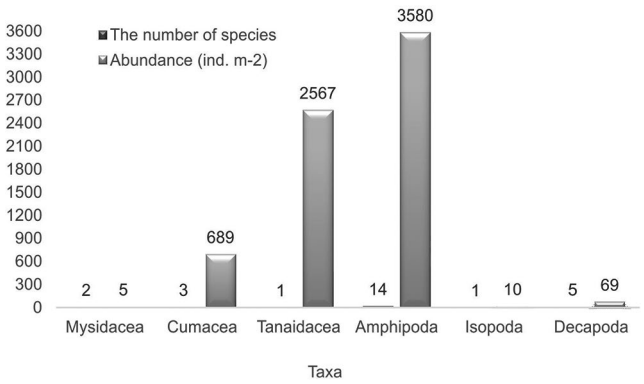


Fig. 4 Crustacea taxa abundances in the study area in the Çardak Lagoon

Organic matter amount in water and sediment

The seasonal distribution of the organic matter amount measured in the water and the sediment was given in Table 3.

Faunistic data

A total of 24 species and 6920 specimens of crustaceans (2 Mysidacea, 3 Cumacea, 1 Tanaidacea, 14 Amphipoda, 1 Isopoda, 5 Decapoda) were recorded in Çardak Lagoon between autumn 2018 and summer 2019. The most dominant taxa in the study area were the amphipods with 3580 ind. m^{-2} followed by Tanaidacea, represented by 2567 individuals within a single species. Amphipod crustaceans were represented by about 51.73% of the total abundance (Fig. 4; Table 4). The most abundant species in the study area was the detritivore tanaid, *Chondrochelia savingyi,* with 2567 individuals and a dominance value of 37.10%. The most dominant species of amphipods were *Microdeutopus gryllotalpa* with 1405 ind. m^{-2} and 20.30% dominance and *Monocorophium acherusicum* with 1062 ind. m^{-2} and 15.35% dominance.

Considering the seasonal distribution of taxa, amphipods and *C. savingyi* were dominant in autumn. The second important sampling period in terms of cumacean, amphipod, and tanaid abundance was winter (Fig. 5).

Among the dominant species, *C. savingyi* predominated in the study area in both sampling periods with a dominance value of 37.10%. The least observed species in the study area were *Leptomysis* sp., *Apherusa* sp., *Philocheras fasciatus*, and *Processa macrophthalma* with a dominance value of 0.01% (Table 4). *Chondrochelia savingyi*, *Ericthonius argenteus*, *Microdeutopus gryllotalpa*, and *Monocorophium acherusicum*, which were observed in the study, were the most abundant species in autumn, at rates of 50%, 100%, 62.41%, and 98.21%, respectively.

Correlations between crustacea abundance and environmental variables

Highest positive correlation ($r = 0.77$; $p < 0.05$) between the environmental variables measured at the stations and abundance was with the level of anionic surfactants in the water. According to the sampling periods, a very high positive correlation value ($r = 0.96$; $p < 0.05$) with $NO_2 + NO_3$ (mg L^{-1}) was found (Table 5).

The correlations between the environmental sediment variable values and the abundance of the dominant species recorded for the sampling periods are presented in Table 6. Table 6 shows that the highest positive correlations were between *Cumella limicola* and *Iphinoe tenella* abundance and % mud content, $r = 0.87$, $r = 0.3$ ($p < 0.05$), respectively. These values were rather weak and negative for the sampling stations. *Ericthonius argenteus*, *Microdeutopus versiculatus*, and *Mononcorophium acherusicum* were the species showing a high positive correlation of $r = 0.81$ ($p < 0.05$) with % sand content in the sediment during the sampling periods (Table 6). Also, the highest negative correlations by station and season for the abundance of the most dominant

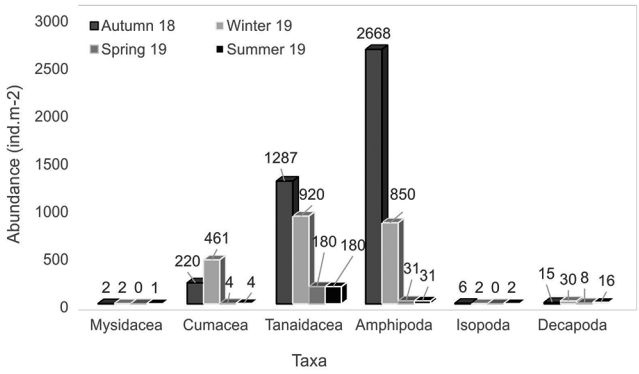


Fig. 5 Crustacea taxa abundances in the Çardak Lagoon in the sampling seasons

Chondrochelia savingyi in the study area were with the pH of the water, r = -0.77 and r = -0.97 ($p < 0.05$), respectively.

TUBI values in sampling stations and in sampling periods were between 2.93 and 3.60 and 2.52–3.47, respectively. According to the TUBI values the study area can be defined as clear in terms of ecological quality (Table 7).

Correlations between biotic indices and environmental variables

Among variables measured for the sampling periods, COD content in water had the highest correlations with H' and TUBI values. ($r = 0.98$, $r = 0.96$; $p < 0.05$, respectively). Moreover, there was a strong correlation between the H' and TUBI values and the average COD (mg L^{-1}) level in water ($r = 0.98$, $r = 0.96$, respectively). Table 8 shows that correlations between TUBI values and the environmental variables such as temperature, salinity, pH, chl-*a*, SiO_2, TP, anionic surfactants in water, gravel %, and OM % in sediment were negative. However, there were strong positive correlations between BENTIX values and O_2 and sand (%) content values (Table 8).

The correlations between the annual mean environmental variable values measured at the sampling points and the biotic indices are presented in Table 9. pH and the sand (%) content in the sediment had the highest correlation with the calculated H' ($r = 0.91$, $r = 0.87$; $p < 0.05$, respectively) and TUBI ($r = 0.76$, $r = 0.79$; $p < 0.05$, respectively) indices within the sampling points. TUBI index had a negative correlation with most of the variables (13 of them).

Correlations with environmental variables of the number of species and abundance

Principal component analysis (PCA) was performed to determine the relationship between environmental variables and the number of crustacean species and abundance. The analysis explained 83% of the model. Anionic surfactant level was the most effective environmental variable in species number and abundance distribution (Fig. 6).

Discussion

Human induced eutrophication in coastal lagoons causes a decrease in water quality and consequently deterioration of natural habitats (Pilo et al. 2011). Declining biodiversity in polluted waters may be a measure of environmental stress on benthic communities (Frontalini et al. 2009). Crustaceans which are bioindicators of polluted waters (Rinderhagen et al. 2000) are very sensitive to high organic matter accumulation in the sediment (Del Valls et al.

Table 4 The number of individuals of crustacean species by seasons and sampling stations in the Çardak Lagoon

Order	Autumn-18	Winter-19	Spring-19	Summer-19	St. 1	St. 2	St. 3	St.4	St. 5	St. 6	St. Ref.	∑	Di%
Mysidacea													
Leptomysis sp.	0	1	0	0	0	0	0	1	0	0	0	1	0.01
Siriella jaltensis Czerniavsky, 1868	2	1	0	1	1	0	0	2	1	0	0	4	0.06
Cumacea													
Cumella limicola Sars, 1879	1	153	0	0	0	15	4	121	4	7	3	154	2.23
Iphinoe tenella G.O. Sars, 1878	219	308	3	3	26	57	84	230	91	42	3	533	7.70
Vaunthompsonia cristata Bate, 1858	0	0	1	1	2	0	0	0	0	0	0	2	0.03
Tanaidacea													
Chondrochelia savignyi (Kroyer, 1842)	1287	920	180	180	1049	214	126	583	343	246	6	2567	37.10
Amphipoda													
Ampelisca sarsi Chevreux, 1888	18	22	0	0	0	14	0	14	0	11	1	40	0.58
Ampithoe ramondi Audouin, 1826	14	49	0	0	0	12	0	39	10	0	2	63	0.91
Apherusa sp.	0	1	0	0	0	0	0	0	0	0	1	1	0.01
Caprella sp.	3	20	0	0	9	0	1	13	0	0	0	23	0.33
Dexamine spinosa (Montagu, 1813)	1	29	0	0	1	7	0	1	0	0	21	30	0.43
Ericthonius argenteus Krapp-Schickel, 1993	331	0	0	0	0	53	0	155	57	66	0	331	4.78
Ericthonius difformis Milne Edwards, 1830	1	45	0	0	0	27	2	13	0	2	2	46	0.66
Gammarus crinicornis Stock, 1966	36	6	0	0	1	0	0	7	34	0	0	42	0.61
Microdeutopus gryllotalpa A. Costa, 1853	877	526	1	1	433	217	34	272	383	66	0	1405	20.30
Microdeutopus versiculatus (Bate. 1856)	172	0	0	0	0	0	0	117	0	55	0	172	2.49
Microdeutopus bifidus Myers, 1977	19	0	0	0	0	0	0	0	0	19	0	19	0.27
Microprotopus maculatus Norman, 1867	153	122	29	29	171	26	29	0	1	85	21	333	4.81
Monocorophium acherusicum (A. Costa, 1853)	1043	19	0	0	79	70	1	359	377	174	2	1062	15.35
Phtisica marina Slabber, 1769	0	11	1	1	13	0	0	0	0	0	0	13	0.19
Isopoda													
Stenosoma capito (Rathke, 1837)	6	2	0	2	0	0	0	6	4	0	0	10	0.14
Decapoda													
Carcinus aestuarii Nardo, 1847	2	1	1	3	0	1	0	6	0	0	0	7	0.10
Diogenes pugilator (Roux, 1829)	2	6	5	11	0	1	0	0	0	0	23	24	0.35
Hippolyte leptocerus (Heller, 1863)	11	22	2	1	0	9	5	22	0	0	0	36	0.52
Philocheras fasciatus (Risso, 1816)	0	0	0	1	0	0	0	0	0	0	1	1	0.01
Processa macrophthalma Nouvel et Holthuis, 1957	0	1	0	0	0	0	0	0	0	0	1	1	0.01
Total number of individulas	4198	2265	223	234	1785	723	286	1961	1305	773	87	**6920**	100.00

∑: Total abundance, Di%: dominance

Table 5 Pearson correlation coefficients between crustacean abundance and environmental variables based on sampling points and periods in the Çardak Lagoon

Sampling stations		Sampling periods		Sampling stations		Sampling periods	
EV	A	EV	A	EV	A	EV	A
WOM (mg L^{-1})	-0.37	OM (%)	-0.53	NO_2+NO_3 (mg L^{-1})	-0.51	TP	-0.71
Mud (%)	0.35	WOM (mg L^{-1})	0.26	TN	-0.01	NO_2+NO_3 (mg L^{-1})	**0.96**
Temperature (°C)	-0.41	Mud (%)	0.43	TSS (mg L^{-1})	-0.45	TN	**0.65**
Salinity (‰)	0.06	Temperature (°C)	-0.51	COD (mg L^{-1})	-0.31	TSS (mg L^{-1})	0.17
O_2 (mg L^{-1})	0.00	Salinity (‰)	-0.22	AD (mg L^{-1})	**0.77**	COD (mg L^{-1})	**0.95**
ORP	0.30	O2 (mg L^{-1})	**0.61**	PO_4 (mg L^{-1})	0.33	AD (mg L^{-1})	-0.68
pH	-0.40	ORP	**0.89**	OM (%)	0.02	PO_4 (mg L^{-1})	-0.52
Chl-*a* (µg L^{-1})	0.30	pH	-0.92	Sand (%)	0.15	Sand (%)	**0.58**
SiO_2	0.29	Chl-*a* (µg L^{-1})	-0.07	Gravel (%)	-0.41	Gravel (%)	-0.95
TP	0.01	SiO_2	-0.28				

OMW: organic matter in water, OM: organic matter in sediment, ORP: oxygen reduction potential, TP: total phosphate, TN: total nitrogen, TSS: total suspended solids, COD: chemical oxygen demand, AD: anionic detergent, EV: environmental variable, A: abundance (ind. m^{-2})

Table 6 Values of Pearson correlation ($p < 0.05$) between abundances of abundant crustacean species and sediment variables in the Çardak Lagoon

	In sampling stations				In sampling periods			
Abundant species	Gravel%	Silt + clay%	OM%	Sand%	Gravel%	Silt + clay%	OM%	Sand%
Ericthonius argenteus	-0.76	-0.26	-0.46	0.66	-0.87	-0.03	-0.04	**0.81**
Microdeutopus gryllotalpa	-0.05	**0.59**	0.17	-0.22	-0.93	0.50	-0.60	0.52
Microdeutopus versiculatus	-0.78	-0.36	-0.39	0.72	-0.87	-0.03	-0.04	**0.81**
Microprotopus maculatus	-0.06	0.21	0.26	-0.04	-0.90	**0.60**	-0.71	0.42
Monocorophium acherusicum	-0.48	0.24	-0.22	0.24	-0.88	-0.01	-0.06	**0.81**
Iphinoe tenella	-0.41	0.06	-0.02	0.26	-0.68	**0.83**	-0.93	0.07
Cumella limicola	-0.53	-0.26	-0.30	0.49	-0.07	**0.87**	-0.91	-0.50
Chondrochelia savignyi	-0.17	0.44	0.25	-0.06	-0.92	**0.55**	-0.65	0.48

Table 7 Values of biotic indices (H', BENTIX, TUBI) and their ecological quality ratio in sampling periods and sampling points in the Çardak Lagoon

Stations	N	A	H'	H' EKO	BENTIX	BENTIX EKO	TUBI	TUBI EKO
St. 1	11	1785	1.68	0.34	4.91	0.82	2.93	0.59
St. 2	13	709	2.69	0.54	4.77	0.79	3.38	0.68
St. 3	9	286	2.04	0.41	**5.11**	0.85	3.18	0.64
St. 4	17	1947	**2.90**	0.58	5.06	0.84	**3.60**	0.72
St. 5	11	1305	2.27	0.45	4.91	0.82	3.22	0.64
St. 6	10	762	2.70	0.54	5.20	0.87	3.55	0.71
St. Ref.	12	86	2.71	0.54	5.00	0.83	3.48	0.70
Seasons	N	A	H'	H' EKO	BENTIX	BENTIX EKO	TUBI	TUBI EKO
Autumn 18	19	4180	**2.56**	0.51	**5.16**	0.86	**3.47**	0.69
Winter 19	20	2243	2.52	0.50	5.00	0.83	3.39	0.68
Spring 19	9	223	1.04	0.21	4.67	0.78	2.52	0.50
Summer 19	12	234	1.29	0.26	5.00	0.83	2.77	0.55

N: Number of species, A: abundance

1998; Guerra-García and García-Gómez 2004). However, organic matter amount in sediment and oxygen level in water are important in explaining the sensitivity of crustaceans (Cabral-Oliverira et al. 2014). Domestic disposals into water may force to migrate the fauna in that area. Causes of animal migrations in eutrophic waters include reduced oxygen level and the changes in salinity and nutrient content in sea water (Ateş et al. 2009). Due to high organic matter accumulation, oxygen level and crustacean abundance in polluted waters is mostly low, except for some detritivore species. Supporting this situation, in this study, we found moderately positive correlations between the abundance of the crustacean and the oxygen content, and very weak and negative correlations with the organic matter level in the sediment. Amphipod crustaceans in particular are susceptible to sewage contamination, which reduces their abundance and diversity. While stressor-tolerant species *Monocorophium* are present in polluted

Table 8 Correlations between biotic indices and average environmental variables for the sampling periods in the Çardak Lagoon (Pearson coefficient correlation, $p < 0.05$)

	Biotic Indices				Biotic Indices		
Variables	H'	BENTIX	TUBI	Variables	H'	BENTIX	TUBI
WOM (mg L^{-1})	0.60	0.61	0.62	TN	0.30	0.21	0.29
Temperature (°C)	-0.64	-0.01	-0.56	TSS (mg L^{-1})	0.50	-0.02	0.43
Salinity (‰)	0.07	-0.48	-0.02	COD (mg L^{-1})	**0.98**	0.69	**0.96**
O_2 (mg L^{-1})	0.60	**0.97**	0.68	AD (mg L^{-1})	-0.79	-0.97	-0.85
ORP	0.64	0.65	0.66	PO^-_4 (mg L^{-1})	-0.47	0.14	-0.38
pH	-1.00	-0.76	-0.99	Sand (%)	0.35	**0.79**	0.43
Chl-*a* (µg L^{-1})	-0.22	0.45	-0.12	Gravel (%)	-0.85	-0.91	-0.89
SiO^-_2	-0.15	0.41	-0.06	Mud (%)	0.66	0.05	0.58
TP	-0.70	-0.14	-0.63	OM (%)	-0.79	-0.28	-0.73
NO^-_2+NO^-_3 (mg L^{-1})	**0.84**	0.54	**0.82**				

WOM: Organic matter in water, OM: organic matter in sediment, ORP: oxygen reduction potential, TP: total phosphate, TN: total nitrogen, TSS: total suspended solids, COD: chemical oxygen demand, AD: anionic detergent

Table 9 Correlations between biotic indices and average environmental variables for the sampling points (Pearson coefficient correlation, $p < 0.05$) in the Çardak Lagoon

	Biotic Indices				Biotic Indices		
Variables	H'	BENTIX	TUBI	Variables	H'	BENTIX	TUBI
WOM (mg L^{-1})	-0.28	-0.27	-0.33	TN	-0.68	0.23	-0.59
Temperature (°C)	0.68	-0.16	0.61	TSS (mg L^{-1})	-0.05	0.74	0.12
Salinity (‰)	-0.05	-0.61	-0.18	COD (mg L^{-1})	-0.01	-0.33	-0.09
O_2 (mg L^{-1})	-0.26	-0.62	-0.38	AD (mg L^{-1})	-0.43	0.20	-0.35
ORP	-0.38	-0.26	-0.40	PO^-_4 (mg L^{-1})	-0.28	0.28	-0.19
pH	**0.91**	0.08	**0.87**	Sand (%)	**0.76**	0.30	**0.79**
Chl-*a* (µg L^{-1})	-0.28	0.49	-0.15	Gravel (%)	-0.57	-0.33	-0.61
SiO^-_2	0.33	-0.69	0.15	Mud (%)	-0.82	-0.14	-0.80
TP	0.08	0.34	0.15	OM (%)	-0.84	0.08	-0.77
NO^-_2+NO^-_3 (mg L^{-1})	-0.32	-0.06	-0.32				

sediments, *Ampelisca* species leave polluted environments immediately (Morris and Keough 2001; de la Ossa-Carretero et al. 2012). *Corophium volutator* (Pallas, 1766), one of the most common species in the Mediterranean lagoons, has been described by Signa et al. (2014) as both a non-selective sediment-feeder and a detritivore. This species was not observed throughout the year in the study area. Further, Karakiri and Nicolaidou (1987) pointed out that amphipods, *Bathyporeira pilosa* Lindström, 1855, *Gammarus locusta* (Linnaeus, 1758), decapods, *Athanas nitescens* (Leach, 1814), *Crangon crangon* (Linnaeus, 1758), and *Liocarcinus depurator* (Linnaeus, 1758) are highly sensitive to hypoxic conditions in aquatic environments. Ateş et al. (2009) also reported that cumacean, *Pseudocuma longicornis* (Bate, 1858), isopod, *Idotea baltica* Pallas, 1772, amphipods, *Caprella rapax* Mayer, 1890, *Dexamine thea* Boeck, 1861, decapods, *Pagurus cuanensis* Bell, 1845, and *Pisidia longimana* (Risso, 1816) were mostly sensitivity to domestic wastes. Same authors indicated that this situation can be explained by the low redox potential that occurs in sediment with pollution. Meanwhile, De la Ossa Carretero et al. (2012) stated that *Autonoe spiniventris* Della Valle, 1893 was the most predominant amphipod species on the bottoms with intense domestic pollution in the open sea.

Regarding organic matter amount, which is determinant in sediment pollution, different reactions of crustaceans to the high organic matter amount consumed by some crustaceans (Pearson and Rosenberg 1978) was studied by Ganesh et al. 2014. Organic matter amount in the sediment is related to the particle size and the bottoms with high organic matter content have muddy character (De Falco et al. 2004; Magni et al. 2004, 2008). Significant changes in sediment characteristics are very decisive in crustacean distribution (Magni et al. 2008). Variables such as organic matter and particle size in sediments of closed lagoons have significant impacts on the distribution of macrozoobenthic communities (Munari et al. 2010). There are many studies regarding correlations between benthic assemblages and accumulation of organic matter in sediments of the Mediterranean lagoons. As an example, Reizopolu et al. (2014) emphasized insignificant correlations between benthos abundance and sediment characters in Grado Marano, Cesina, and Logarou lagoons (Adriatic Sea). *Monocorophium*, *Erichthonius*, and *Chondrochelia* species were abundant on the polluted

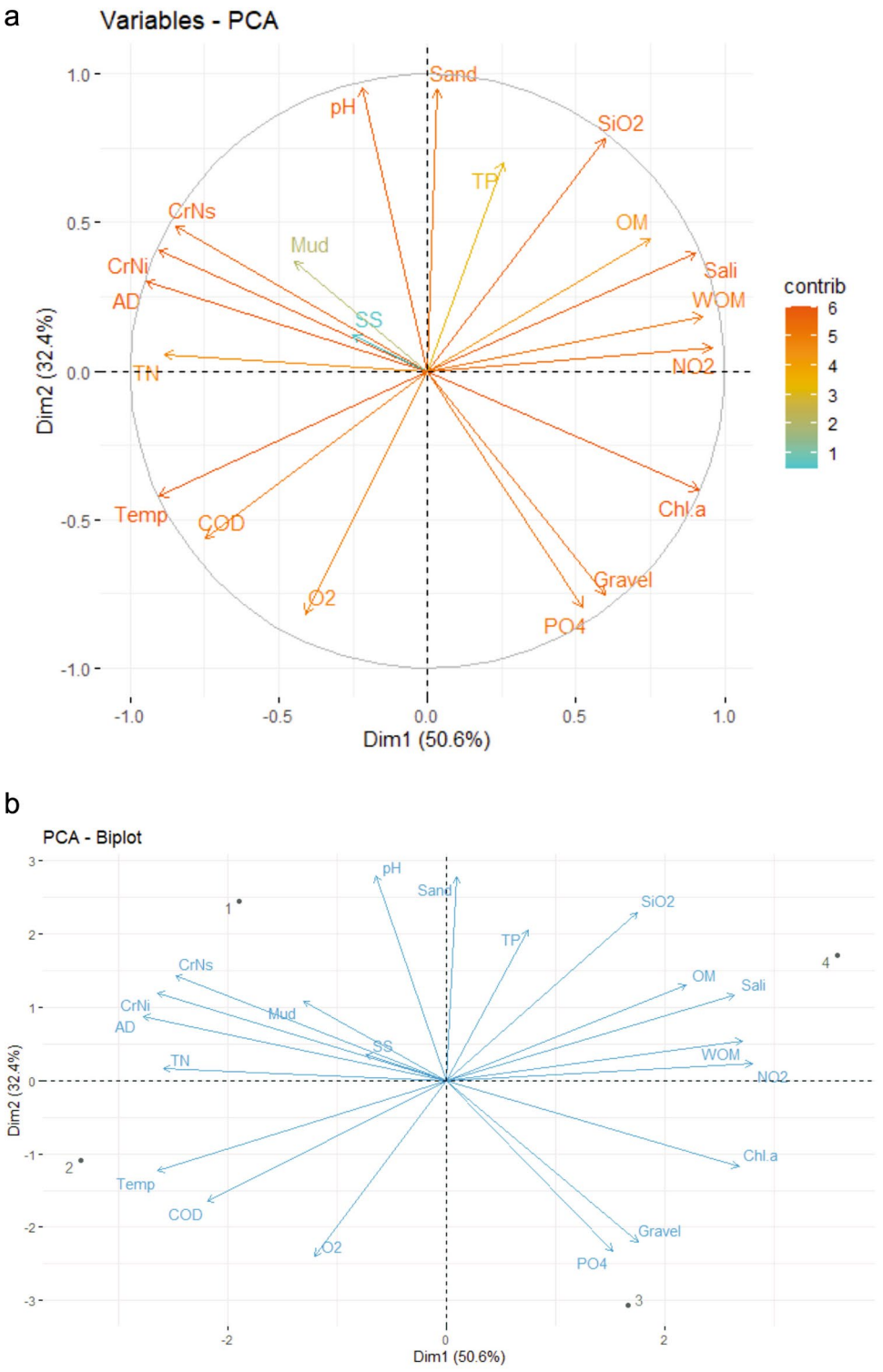


Fig. 6 PCA ordination diagram of the temporal and spatial mean environmental variables and abundance (**a**) and (**b**) species number in the Çardak Lagoon

sediments. Additionally, Chintiroglou et al. (2004) reported that tanaid, *Chondrochelia savingyi* was predominant in the sediments with rich organic matter. Regarding the abundance of *Microdeutopus gryllotalpa*, a detrivitore species that is sensitive to sediment pollution, Drake and Arias (1997) stated that amphipod, *M. gryllotalpa* was dominant on the bottoms with 8.9% organic matter amount in Cadiz Lagoon (southern Spain). In this study, *M. gryllotalpa* was recorded mostly on the sediments with a relatively high organic matter content (7.76%). Furthermore, Lardicci and

Rossi (1998) indicated that the sediments with low organic matter content (2.9–5.13%) of Orbetello Lagoon (western Italy) was dominated by *Monocorophium insidiosum* (Crawford, 1937). The most important lagoon in the Mediterranean, dominated by *Monocorophium* species, especially *M. orientale*, is Papapouli Lagoon (Paraskevopoulou et al. 2015) with low organic carbon contents (1.15–2.81%) in the sediment.

The Amphipod *Microdeutopus gryllotalpa* was abundant in Çardak Lagoon on the bottoms with more than 10% organic matter content where *Chondrochelia savignyi* individuals were dominant on the bottoms with lower organic matter content (4.38%). Contrarily, Evagelopoulos et al. (2008) found *M. gryllotalpa* mostly on bottoms with low organic matter content (1.2–2.04%) in autumn in Kalloni Bay (Lesvos Island, Greece). In autumn *M. gryllotalpa* and *L. savignyi* dominated the study area in sampling points with high nitrogen concentrations. Ponti and Abbiati (2004) noted that the abundance and temporal distribution of detritivore amphipods, *Monocorophium acherusicum* and *M. insidisum* strongly decreases when concentrations of chl-*a* and nutrients in water are high. Yet, we found irregularities between *M. acherusicum* abundance and the average amount of chl-*a* depending on sampling stations. *Monocorophium acherusicum* abundances were high, while chl-*a* concentration in water was on average 2.78 µg L^{-1}. Thus, we found a weak positive correlation between chl-*a* concentration and abundance. In addition, *M. acherusicum* and *Ampithoe ramondi* were the most dominant species in sediments with an average organic matter of 3.49% in the study area. We also observed that *C. savingyi*, *M. acherusicum*, and *M. gryllotalpa* were dominant on the sea-side bottoms of the study area with low organic matter amount (2.75 and 3.49%). On the seaside of Çardak Lagoon, the dominant species were amphipods, *Ampelisca sarsi*, *A. ramondi*, *Ericthonius argenteus*, *E. difformis*, cumacean, *Cumella limicola*, and decapod, *Hippolyte leptocerus*. *Ericthonius argenteus*, *Microdeutopus bifidus*, and *Hippolyte leptocerus* which are highly sensitive to pollution, were also most abundant on the bottoms with low organic matter content (2.75% and 3.49%) of the study area. Our results showed that organic matter amount (%) in the sediment was determinative on the crustacean abundance.

Sediment particle size is one of the most important environmental variables affecting the distribution of crustacean assemblages (Reizopoulou and Nicolaidou 2004; Afli et al. 2008). As particle size decreases in the sediment, the number of species increases, and correlations become weaker (Nicolaidou et al. 1988). Thus, several isopod and amphipod species prefer the sediments with coarse particle size (Afli et al. 2008). For example, Reizopoulou and Nicolaidou (2004) found the highest correlation between the coarse sediment particles and the number of species (r_s = 0.371; $p = 0.002$) in Tsopeli, Tsoukalio, Rodia, and Logarou Lagoons in Amvrakilos Bay (western Greece). McArthur et al. (2000) reported that there was a negative relationship ($\rho = -0.51$; $p < 0.01$) between mud (silt + clay) content % and the abundance of all zoobenthos in Gialova Lagoon (southwestern Greece). Conversely, we recorded high crustacean abundance on the sediments with high (13.24–16.71%) mud content. By species, Magni et al. (2004) represented that *Monocorophium sextonae* (Crawford, 1937) found in Cabras Lagoon showed positive correlations with sediment particle size. Similarly, we found a very remarkable correlation ($r = 0.98$; $p < 0.05$) between *Monocrophium acherusicum* abundance and mud content % in Autumn and Winter periods. We also found a high positive correlation between *M. acherusicum* abundance and organic matter content in Autumn. Furthermore, pollution indicator, *Chondrochelia savingyi* was quite dominant on the bottoms with high mud content in the study area.

Saturated oxygen concentration in the water can fall below 50% during the night period, especially in summer months when nutrient and chl.-*a* concentrations are at high levels. If nutrient concentrations are high, the abundance and temporal distribution of amphipods such as *M. acherusicum* and *M. insidiosum* is strongly reduced due to increased ammonium and phosphate accumulation (Ponti and Abbiati 2004). *Monocorophium acherusicum* specimens were not observed in the summer sampling of this study. The species were mostly recorded during the daytime in autumn at stations where the average oxygen level was 6.86 mg L^{-1} and the average chl-*a* concentration was 2.08 µg L^{-1}. Spatial distribution of amphipod species is affected by climatic changes such as temperature (Jelassi et al. 2015). As an example, Ponti et al. (2011) defined that Pialassa Baiona Lagoon surface water temperature (r = -0.55; $p < 0.001$) and organic carbon content in sediment (r = -0.386; $p = 0.014$) were inversely related to species diversity. In this study, the correlations between crustacean abundance and lagoon water surface temperature were negative.

The effects of domestic pollution in Çanakkale Strait on crustacean assemblages were previously reported by Ateş et al. (2009). Ateş et al. (2009) indicated that the abundance of the dominant species such as *C. savignyi*, *M. acherusicum*, and *Microdeutopus gryllotalpa* did not show statistically significant correlations with the water and sediment variables measured in polluted shallow waters. When the correlations between abundances of 3 species cited above and environmental variables based on seasons and stations were evaluated, irregularities were observed. *Chondrochelia savingyi* abundance showed high correlations with environmental variables depending on the sampling periods and stations (correlation of $r = 0.99$ with chemical

oxygen demand amount for sampling periods and r = 0.88 with anionic surfactant concentration for sampling stations). *Monocorophium acherusicum* abundance had strong correlations with seasonal redox potential and anionic surfactant concentration (relatively r = 1.00 and r = 0.87; $p < 0.05$), and weak correlations with all variables with respect to stations. *Microdeutopus gryllotalpa* abundance was positively correlated with 9 different environmental variables, and the highest correlations close to 1 were with O_2 (r = 0.99; $p < 0.05$), $NO_2 + NO_3$ (r = 0.98; $p < 0.05$), and COD (r = 0.96; $p < 0.05$) in water. It has been thought that mud and sand content % in sediment were the two important environmental factors in determining the ecological quality status (for H′ and TUBI values) of Çardak Lagoon.

The water column and sediment of the land side of the study area was slightly affected by domestic wastes originating from Çardak Town and dairy facilities nearby. Although pollution concentrations in lagoon ecosystems were low (example of annual average of 15.52% organic matter content in sediment at two of the landside sampling stations of the lagoon), COD values may be high due to organic matter accumulation. Transport or dilution of the pollution in shallow lagoons is most likely possible with wave movements caused by rains and strong winds. Small wave movements by weak winds in the lagoon for autumn and summer may cause the accumulation of organic matter in the sediment from domestic pollution. The strong south and north winds observed through the year in the study area reduced the effect of pollutants in the lagoon area.

Conclusion

Considering the abundance of crustaceans, anionic surfactant level for sampling points, COD, $NO_2 + NO_3$ and ORP levels are effective on crustacean abundance during the sampling periods. The results of this study partially supported the findings on the distribution patterns of crustaceans and the relationships between abundance and environmental variables reported in various lagoons of the Mediterranean. Çardak Lagoon, which is slightly closed, seemed to have been less affected by the anthropogenic effects. The low water regeneration rate and domestic wastes did not seem to negatively affect the crustacean community structure. It is clear that Çardak Lagoon is exposed to many environmental stressors that have an impact on the existing crustacean community structure and responds to stress due to the low densities of communities, especially at coastal sampling points where sediment pollution was observed. Thus, further study on the analysis of crustacean communities with environmental variables could contribute to having more insight into ecosystem functioning.

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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